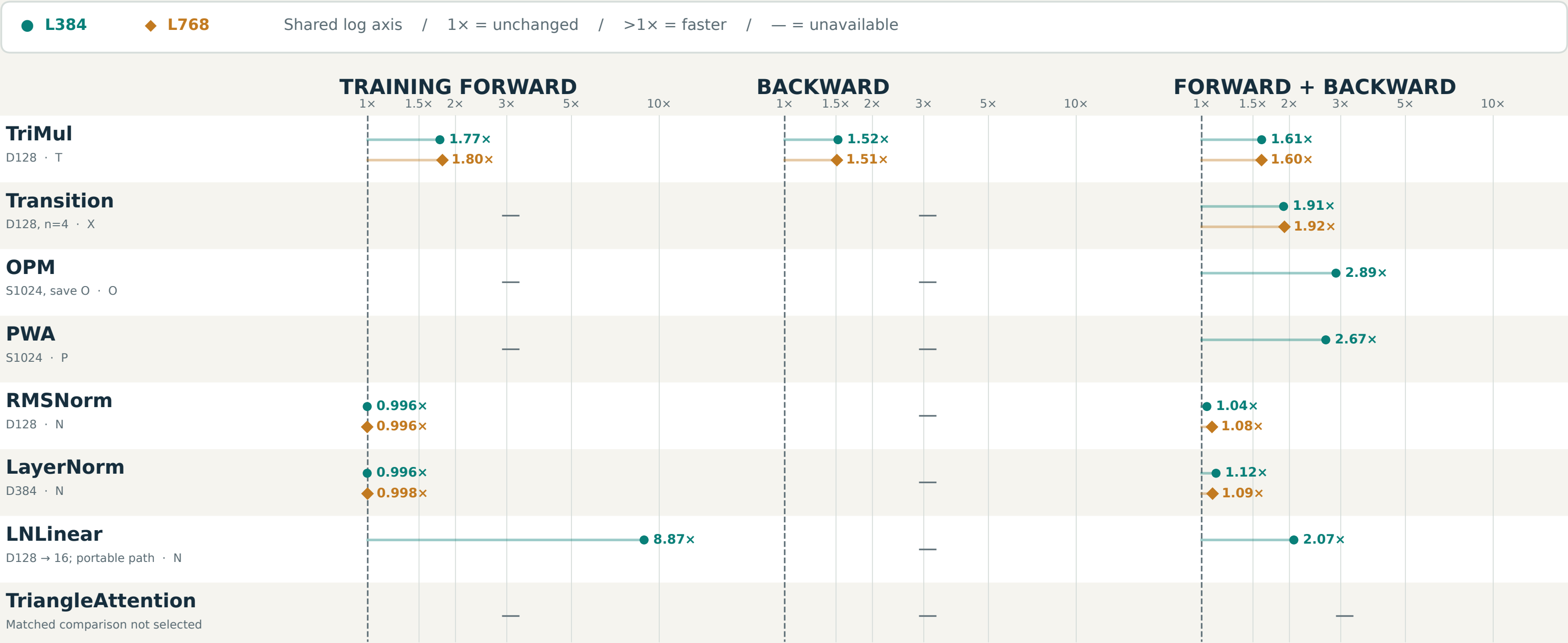


# Measure the full operator, not just the fast stage

Historical H100 records • speedup = baseline time / our time • each source keeps its own baseline • no new GPU measurements



T existing Triton X Triton residual O/P own engine MSA paths N pre-change portable Triton  
Forward means training forward here. F+B is measured directly; backward is never inferred by subtraction. No aggregate speedup is computed.

## TRIMUL WIDTH SCALING

Research checkpoints for D256+ are not promoted to shared dispatch.  
Teal ≥1.5x; blue <1.5x.

|      | L384 BWD | L384 F+B | L768 BWD | L768 F+B |
|------|----------|----------|----------|----------|
| D128 | 1.52x    | 1.61x    | 1.51x    | 1.60x    |
| D256 | 1.27x    | 1.35x    | 1.70x    | 1.66x    |
| D384 | 1.39x    | 1.39x    | 1.53x    | 1.51x    |
| D512 | 1.32x    | 1.35x    | 1.50x    | 1.51x    |